

ECSE 2500
Feb. 9th
Lee 9

2/13

Monday

Review of problems
related HW1-3/Exam

TA Guo

2/16

Thursday

Cancelled

2/21

Monday $\rightarrow$ Tuesday

Exam 1

One-page sheet

Today's topic: Expected value and moments of RV.

Last time, we talked about discrete random variables; we will perform an expectation of that random variable.

Q: What is the key component of an RV?

A: Probability Mass Function (PMF): tells us the probability of a random variable taking a particular value.

Need a table to store PMF

s	$X(s)$	$P(X = X(s))$

o The **Expected Value** of a (discrete) random variable (aka. Expectation or Mean) is defined as the sum of all possible values the RV X can take weighted by their probabilities:

$$E[X] = \sum_{x_k \in S_X} x_k P_X(x_k)$$

$\uparrow$
 $P(X=x_k)$

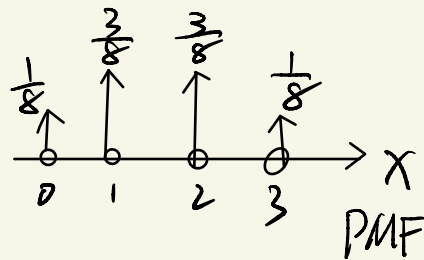
Example: $X = \#$ of heads in 3 flips of fair coin

$P_X(x_k) = \text{Probability } X = x_k$

$$E[X]$$

$$= 0 \times \frac{1}{8} + 1 \times \frac{3}{8} + 2 \times \frac{3}{8} + 3 \times \frac{1}{8}$$

$$= \frac{3}{2} = 1.5$$



Makes sense intuitively since PMF is symmetric around this value 1.5.

Note: $E[X]$ is not the same as the most likely value X takes; And most of the times, it is not actually an outcome of this experiment (e.g., We cannot get 1.5 heads)

In fact, the intuition is that $E[X]$ is the limit of sample mean: If do a large number of Experiments and record outcomes, we have

$$\text{Sample mean} = \frac{\sum_{X_k=0}^{\infty} \# \text{ of trials that we see } X_k \text{ heads}}{\text{Total \# of trial}} X_k$$

As $\#$ of trials $\rightarrow \infty$, Sample mean $\rightarrow E[X]$.

Example

1) X is a Bernoulli RV, $P_X(0) = 1-p$

$$P_X(1) = p.$$

$$E[X] = 0 \times (1-p) + 1 \times p = p$$

e.g., Say I get one dollar every time a trial succeeds. My expected payoff is $p \cdot$ one dollar.

2) X is a Geometric RV,

$$P_X(k) = (1-p)^{k-1} \cdot p, \quad k=1, 2, \dots, \infty$$

$$E[X] = \sum_{k=1}^{\infty} k P_X(k)$$

$$= p \sum_{k=1}^{\infty} k (1-p)^{k-1}$$

$$= p \cdot \left(\frac{1}{1-(1-p)} \right)^2 = \frac{1}{p}$$

How to simplify?

$$\sum_{k=1}^{\infty} k a^{k-1} \quad a \in (0, 1)$$

$$= 1 + 2a + 3a^2 + 4a^3 + \dots$$

$$= 1 + a + a^2 + a^3 + \dots$$

$$+ a + a^2 + a^3 + \dots$$

$$+ a^2 + a^3 + \dots$$

$$\begin{array}{r} \frac{1}{1-a} \\ \frac{a}{1-a} \\ \frac{a^2}{1-a} \end{array}$$

$$= (1 + a + a^2 + \dots) \frac{1}{1-a} = \frac{1}{(1-a)^2}$$

3) X is Poisson RV with parameter λ

$$P_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}$$

$$E[X] = \sum_{k=0}^{\infty} k \cdot \frac{\lambda^k}{k!} e^{-\lambda}$$

$$= e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{(k-1)!}$$

$$= e^{-\lambda} \cdot \lambda \sum_{k=0}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}$$

$$= \lambda \cdot e^{-\lambda} \cdot e^{\lambda}$$

$$= \lambda$$

↑
Taylor series for e^{λ}
 $e^{\lambda} = 1 + \lambda + \frac{1}{2!} \lambda^2 + \frac{1}{3!} \lambda^3 + \dots$

Which means that the expected value of a Poisson RV is the parameter λ .

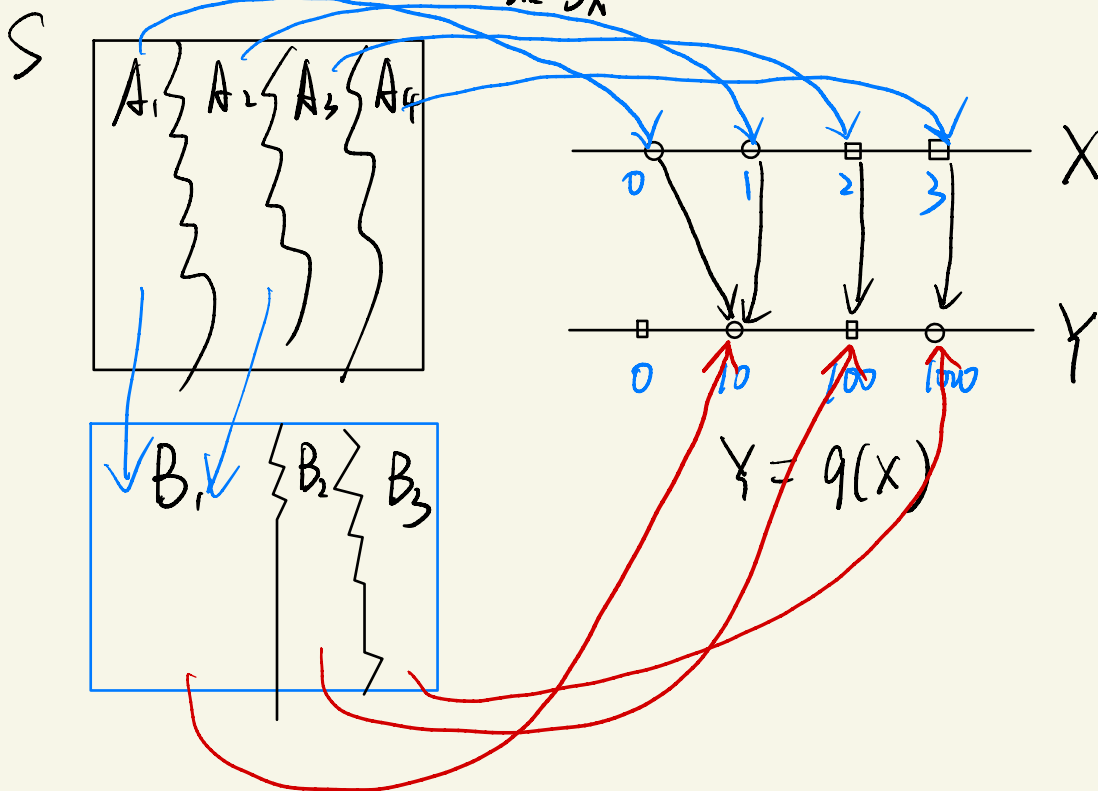
□ Expected Value of A function of RV X

$$E[g(x)].$$

$\uparrow \quad \uparrow$
 $R \rightarrow R \quad S \rightarrow R$

Definition :

$$E[g(x)] = \sum_{x_k \in S_X} g(x_k) \cdot P_X(x_k)$$



Based on the alternative view of partition sample space,

$$E[g(X)] = E[Y]$$

$$= \sum_{y_k \in S_Y} y_k P_Y(y_k)$$

$$= \sum_{y_k \in S_Y} y_k \left(\sum_{x_j \in S_X | g(x_j) = y_k} P_X(x_j) \right)$$

$$= \sum_{x_j \in S_X} g(x_j) P_X(x_j) = E[g(X)]$$

Some important properties

$$E[g(X) + h(X)] = E[g(X)] + E[h(X)]$$

$$E[aX] = a E[X]$$

$$E[X + c] = E[X] + c$$

$$E[c] = c$$

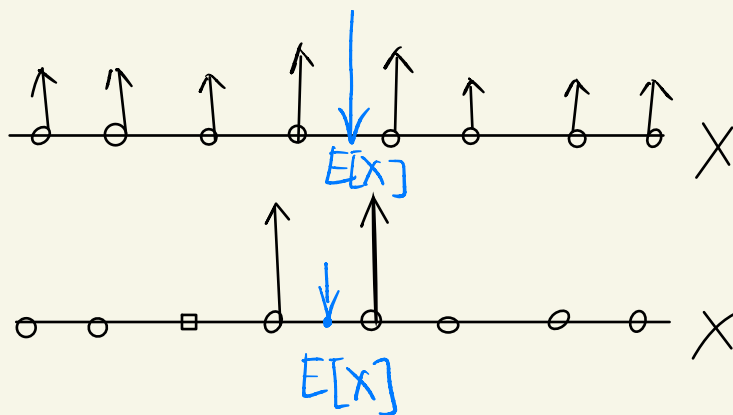
□ Variance — A special case of Expected value of a function of random variable

Define a function

$$g(X) := (X - E[X])^2$$

e.g. How far do we expect X to be from $E[X]$?

Give us some sense of the spread of PMF.



Variance

$$\begin{aligned}\text{Var}(X) &= E[g(X)] = E[(X - E[X])^2] \\ &= E[X^2 - 2XE[X] + E^2[X]] \\ &= E[X^2] - 2E[X]E[X] + (E[X])^2 \\ &= E[X^2] - (E[X])^2 \leftarrow \text{Important}\end{aligned}$$

We often use $\text{Var}(X) = E[(X - E[X])^2]$

and $\sigma_X = \sqrt{\text{Var}(X)}$, aka. standard deviation

Bigger $\sigma_X \Rightarrow$ Values of RV widely spread around the mean of RV.

Moment of X

$E[X^k] \leftarrow k\text{th moment}$

Central moment of X

$E[(X - E[X])^k]$

$\leftarrow k\text{th central moment}$

Example: 1) $X \sim \text{Binomial RV}$, what is $E[X]$?

$$P_X(k) = C_k^n p^k (1-p)^{n-k}, \quad k = 0, 1, 2, \dots, n$$

$$E[X] = \sum_{k=0}^n k C_k^n p^k (1-p)^{n-k} \quad \text{Looks a pain}$$

We can view Binomial RV X

$$X = \underbrace{X_1 + X_2 + \dots + X_n}_{n \text{ independent Bernoulli RV}}$$

$$E[X] = \sum_{k=1}^n E[X_k] = n p$$

2) Variance of a Bernoulli RV with parameter p .

$$E[X] = p.$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$= 0^2 \cdot (1-p) + 1^2 \cdot p - p^2 = p - p^2$$

Important property

If X and Y are two independent RVs, their variance

$$\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y)$$